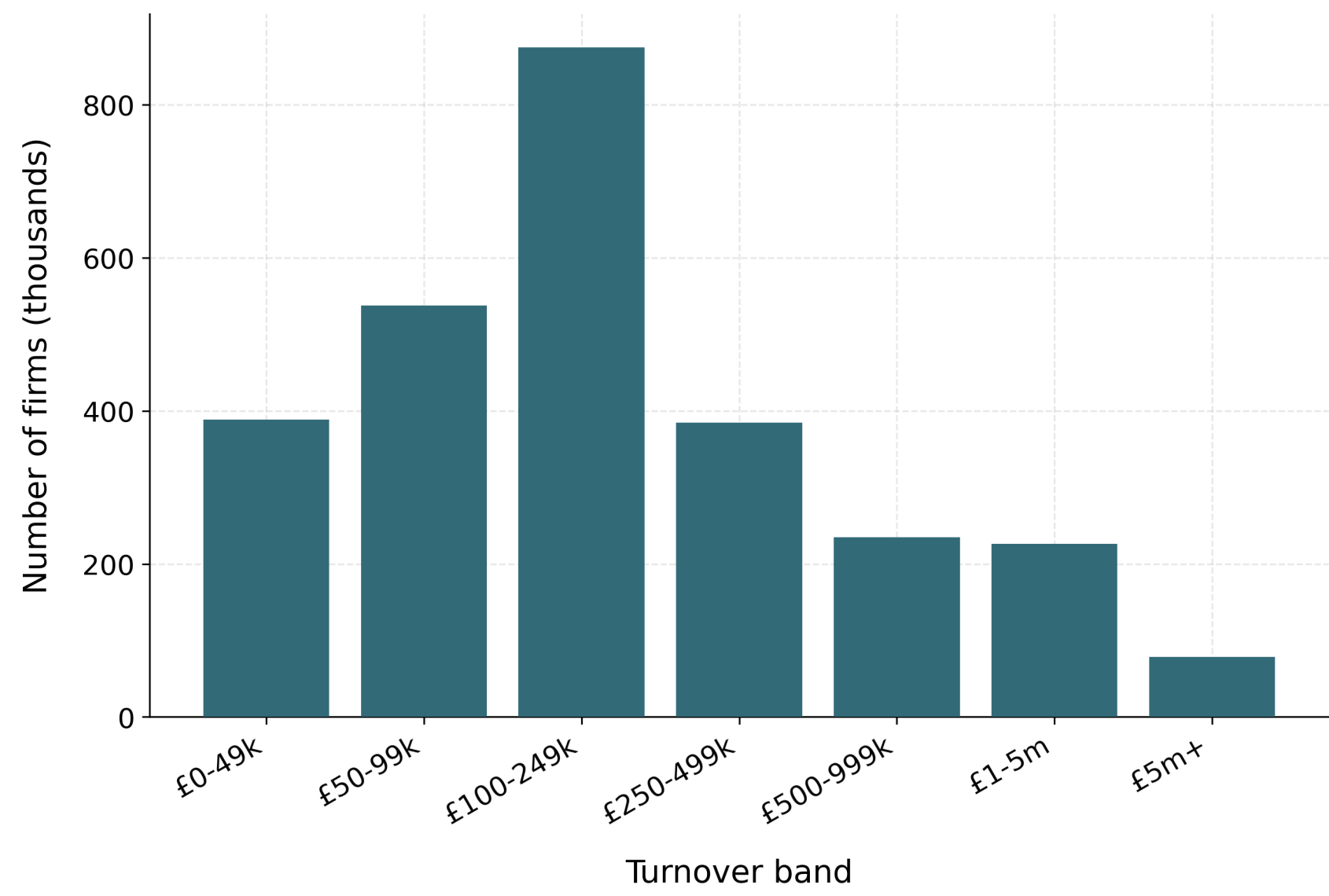


The problem: VAT as a tax *notch*

The UK VAT registration threshold is a **notch**, not a **kink**. Once taxable turnover crosses T^* , **20% VAT falls due on the *whole* turnover**, not just the increment — a discrete liability jump of $\tau T^* = \$17,000$ at $T^* = \$85,000$. Threshold **£85,000** (2023–24 data year), raised to **£90,000** in April 2024 — among the **highest in the OECD** — and it sits in a *dense* part of the firm-size distribution.



Contribution

Headline: only the graduated taper removes the notch's distortion; the level move is the **dearest and least efficient**.

An open, reproducible firm-level microsimulation that:

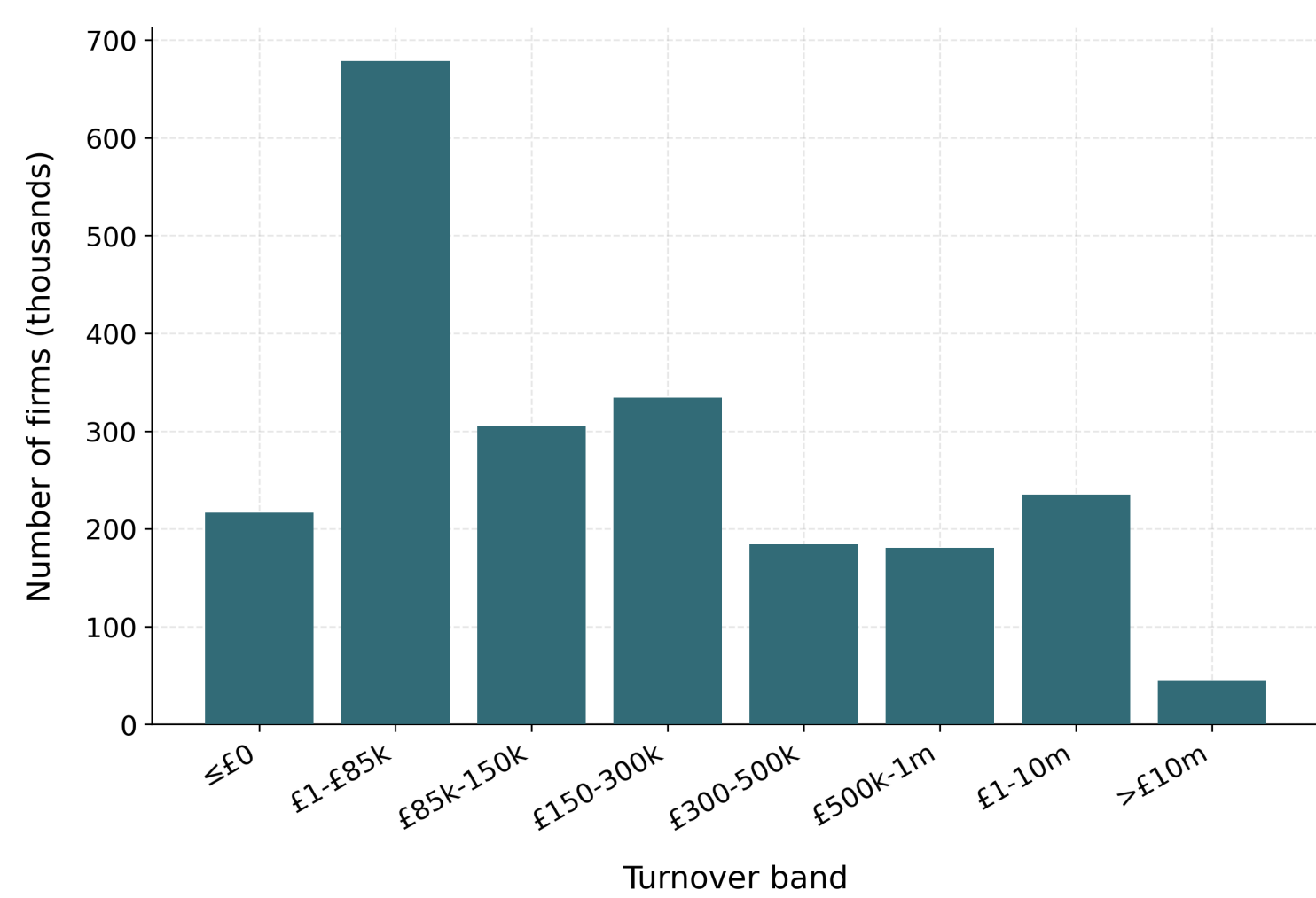
- costs **level + shape + rate** reforms on one static basis;
- characterises the notch's distortion via its exact, **elasticity-free dominated region**;
- shows via a **placebo** that synthetic bunching is mechanical;
- adds a conditional **behavioural layer**.

Three reform families, one basis

What we cost.

Costed on a *common static basis*:

- **Level** — move the threshold up or down.
- **Shape** — a graduated taper that phases VAT in.
- **Rate** — a reduced-rate band above the threshold.



Method in one line

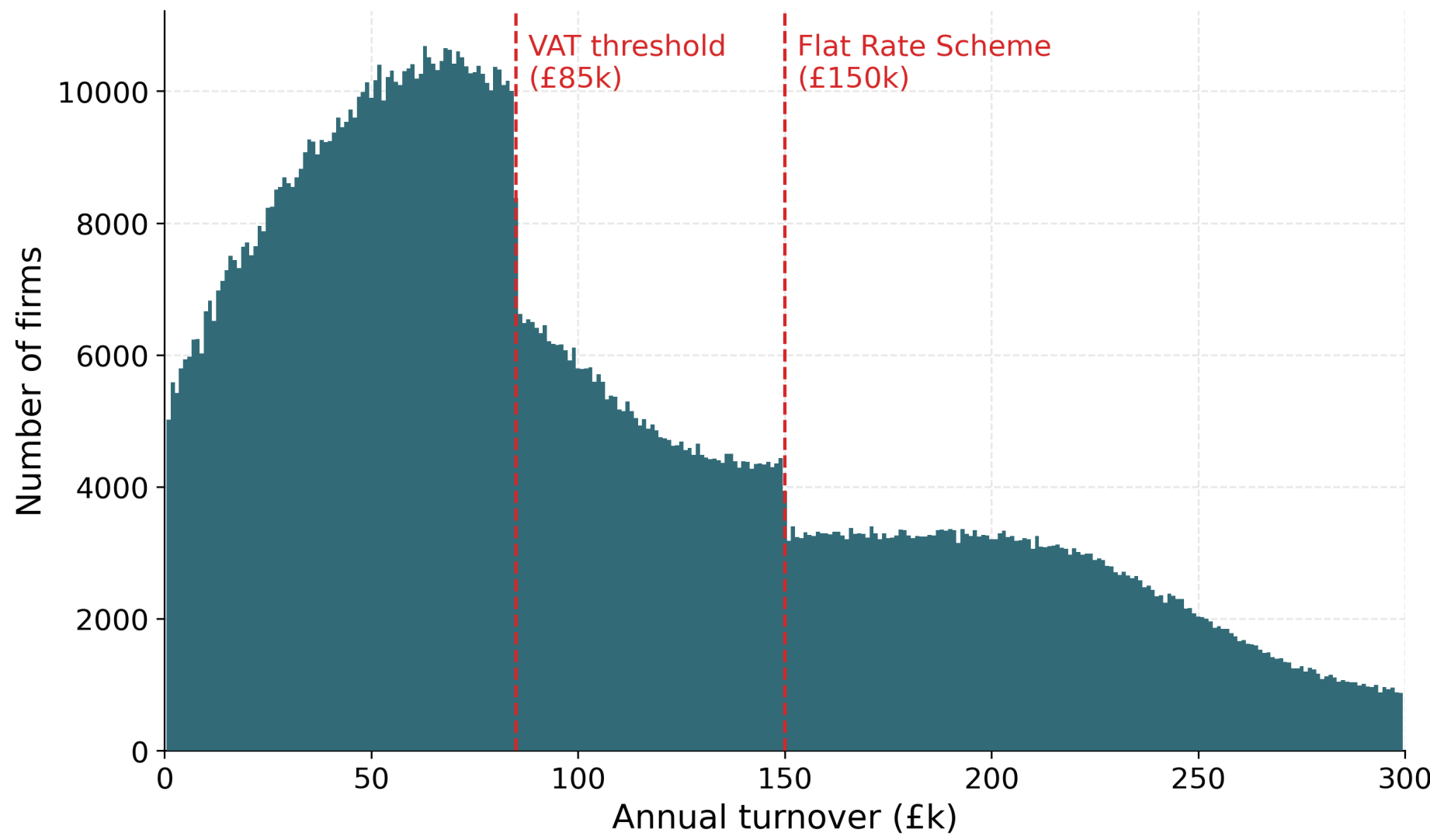
How the synthetic data are built.

Draw synthetic firms by sector × turnover band; assign value added $v_i = y_i - x_i$. The threshold is a *generator parameter*, so data regenerate under any counterfactual. Firm weights $w_i = e^{\theta_i}$ are fit by gradient descent to HMRC + ONS aggregates.

Synthetic firm data

→ The engine behind every costing.

≈2.94m firm rows weighted to ≈2.5m UK firms, calibrated to HMRC + ONS aggregates (turnover-band counts, sector, employment, net VAT liability by band) at ≈90% accuracy. Calibrated to the aggregates, so agreement is an **internal-consistency check**, not external validation.



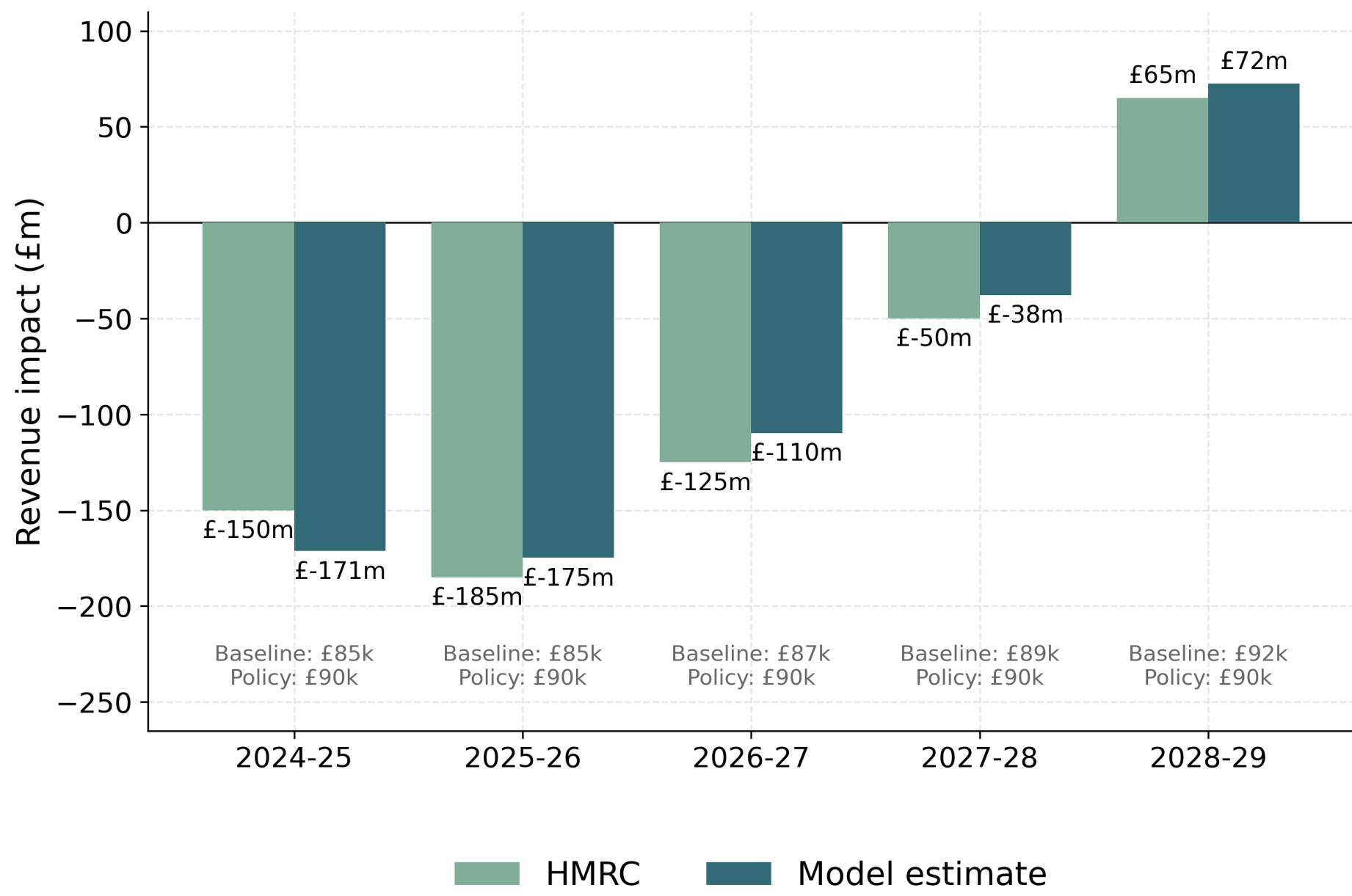
Static costing

Pricing a reform on this data.

$$R(T) = \sum_i w_i v_i \mathbb{1}\{y_i \geq T\}$$

Revenue moves only through firms whose registration status flips.

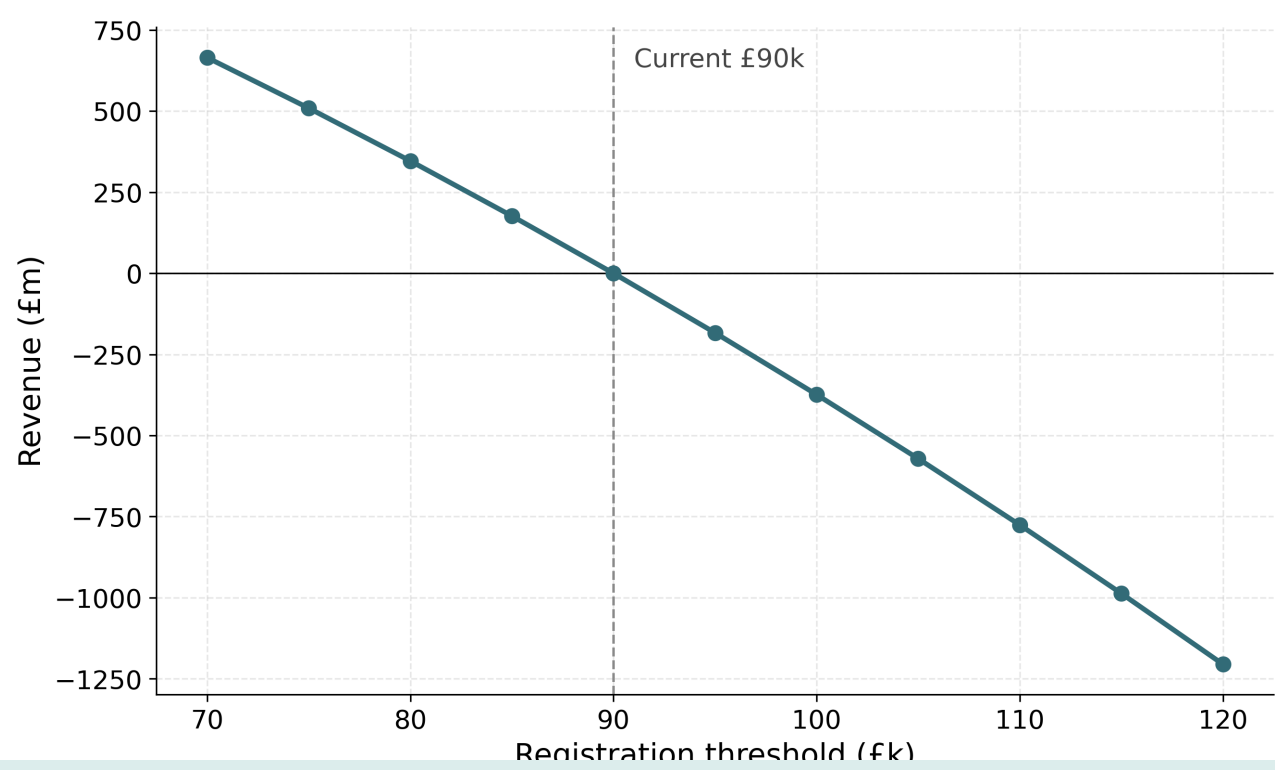
Validation vs HMRC: the £85k→£90k April-2024 increase costs **–£175m** in 2025–26 vs HMRC's published **–£185m** — agreement within **~£10m**.



Threshold sweep (2025–26)

Scaling the level reform across thresholds.

From a £90k base (**£203.8bn**), the relationship is roughly **linear**, ≈£150–220m per £5k step.



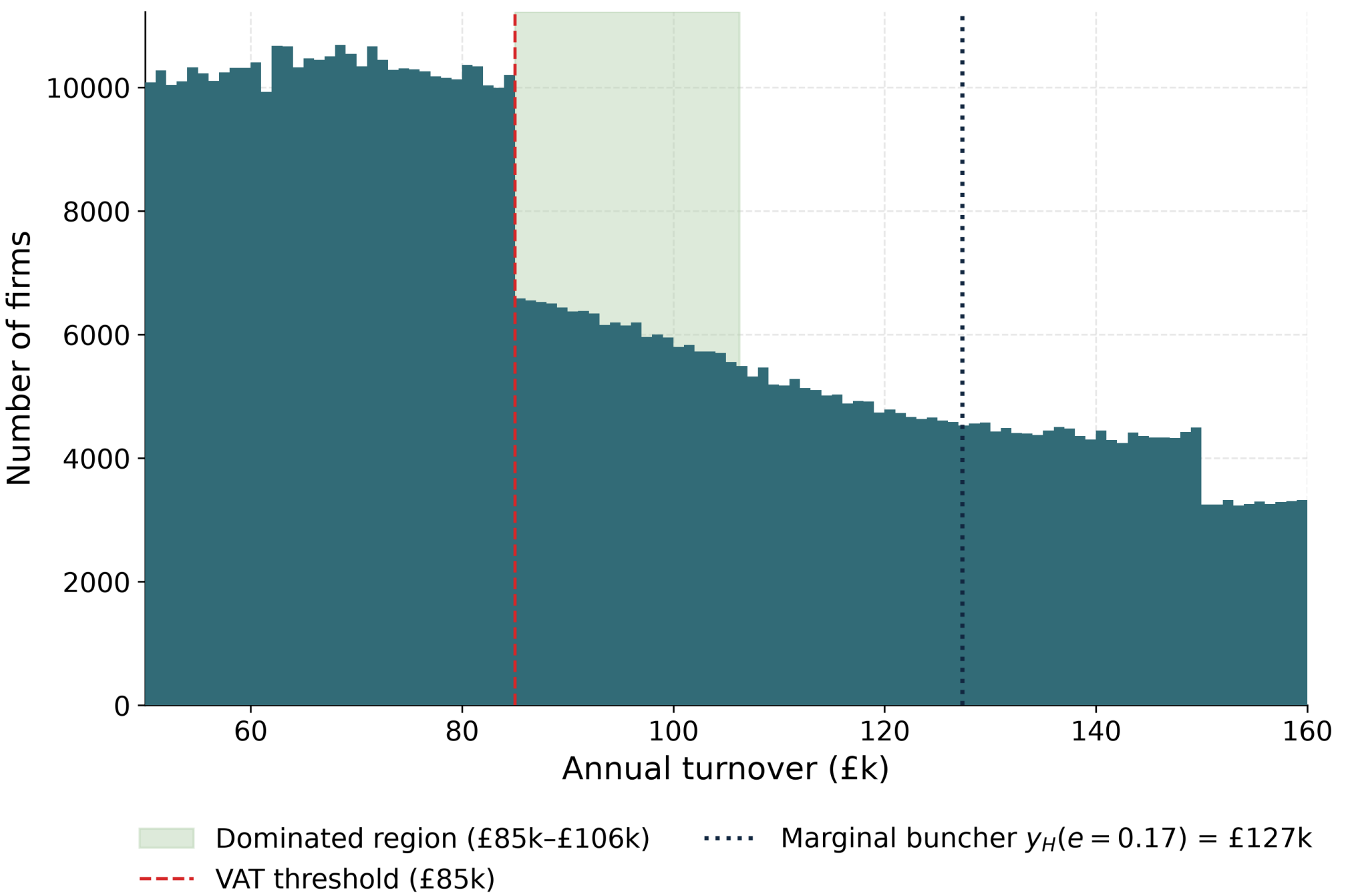
The dominated region

→ Beyond revenue: measuring the notch's distortion.

Profit jumps *down* by $\tau T^* = \$17,000$ at the threshold. The **dominated region** — turnover where no firm locates because it earns less than at T^* — has exact width

$$a = T^* \frac{\tau}{1-\tau} = \$21,250$$

(£85,000–£106,250), spanning ≈137,000 weighted firms. **Exact and elasticity-free** (conditional on the turnover-tax notch).



How reforms change it

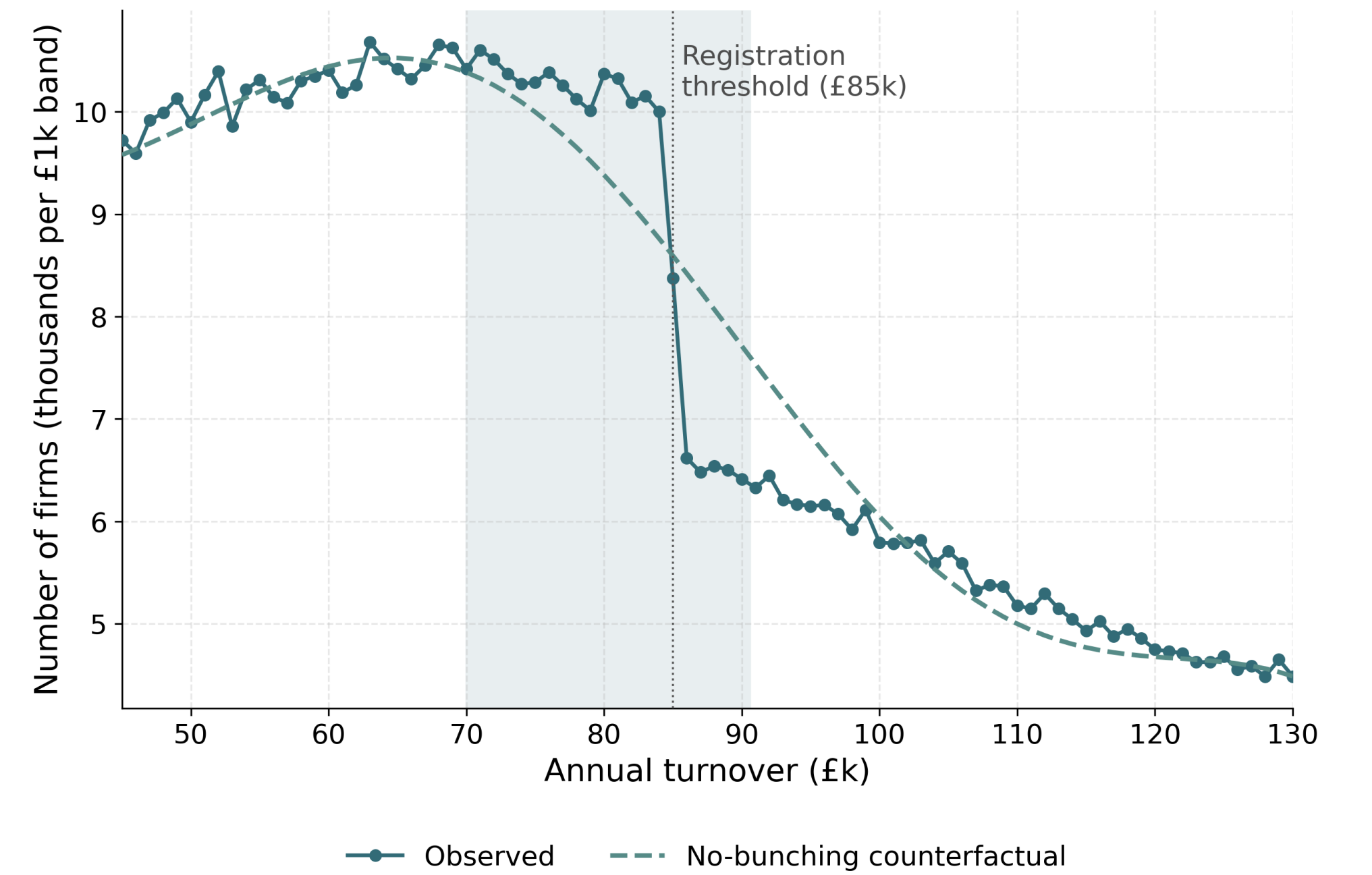
Which reforms actually shrink this band?

- Raising the threshold **relocates** the £21,250 band unchanged.
- A banded reduced rate does **not shrink** it: reverting to 20% at the band top adds a *secondary notch* $a' = T_1 \frac{\tau}{1-\tau}$, so total width is **£21,562 (15%) / £22,569 (10%)** — it **splits** the band, not shrinks it.
- **Only the graduated taper removes the distortion entirely.**

Bunching is mechanical

A transparency check on the synthetic data.

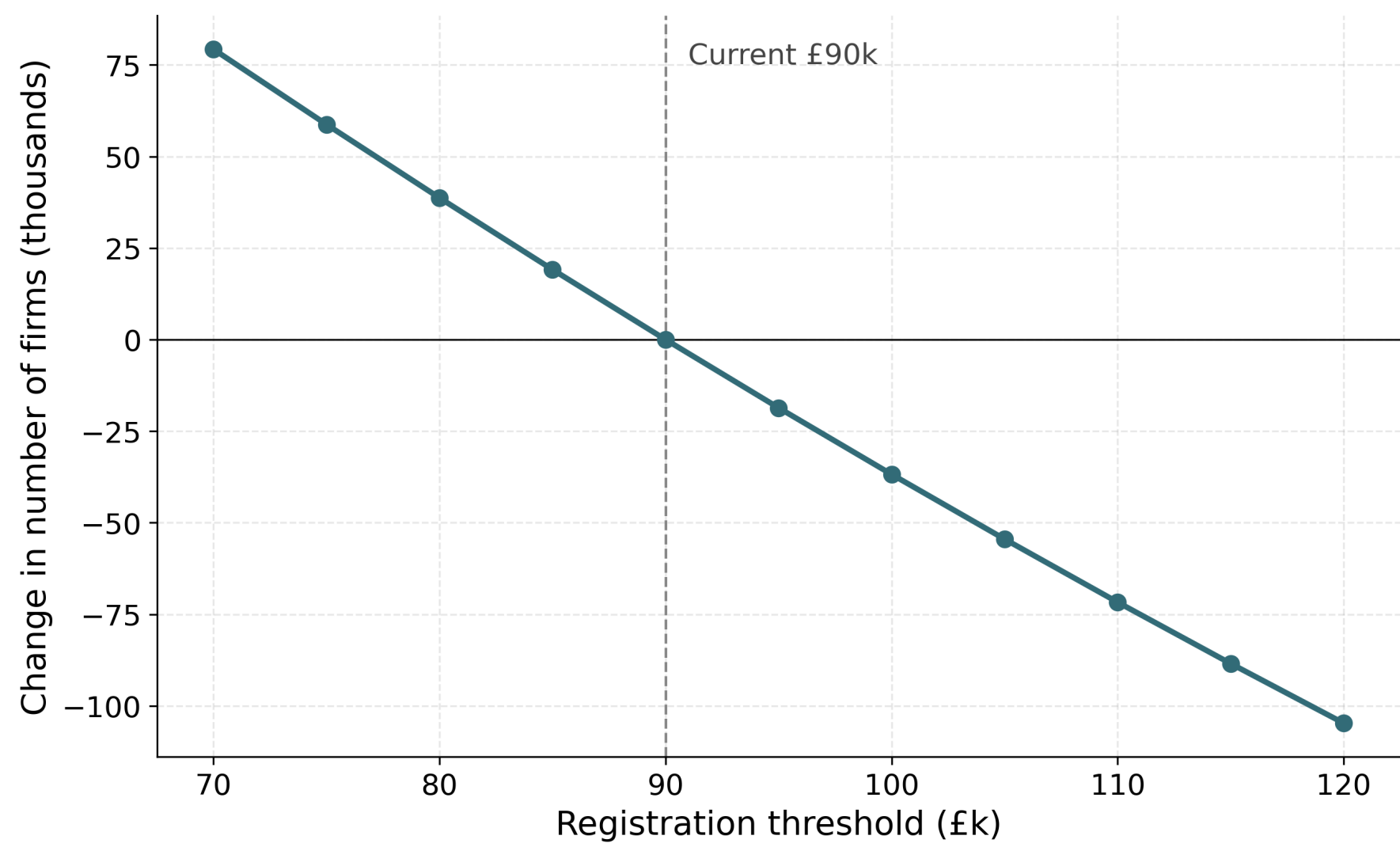
Estimator: $b = 0.060$, excess mass $E = 8,712$ firms. A **placebo** collapses E to **0–99** when the step is removed from the calibration target ⇒ bunching is inherited from coarse HMRC band targets, *not* firm behaviour. Behavioural bunching is therefore **not identified** on the synthetic data; the administrative estimate of Liu et al. (2021) remains the behavioural fact.



Reform menu (common £85k / £183.6bn base)

→ Cost and distortion, side by side.

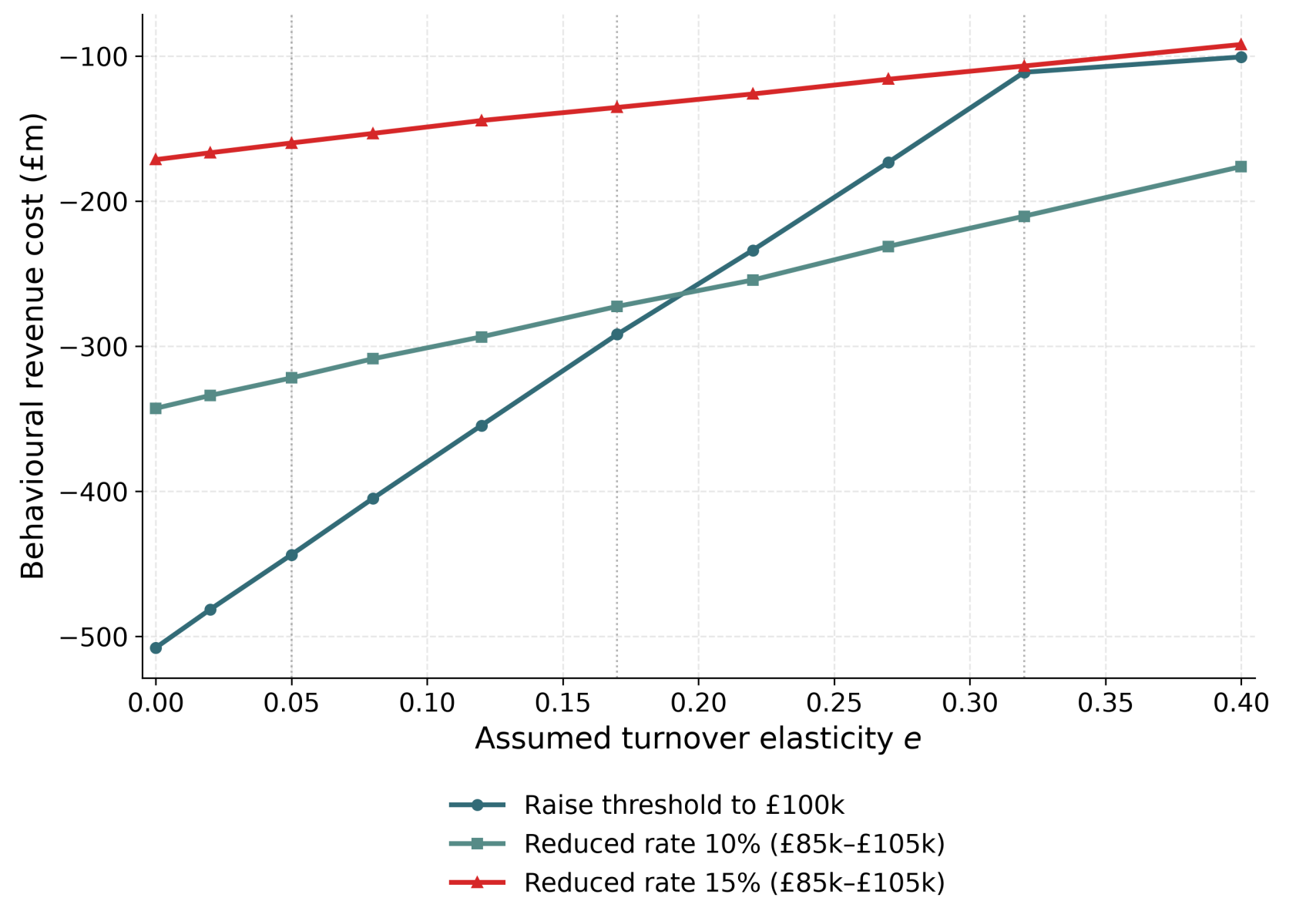
Reform	Type	Static	Dominated region
Raise to £100k	Location	–£508m	relocates, removes none
Taper £85–105k	Shape	–£336m	removed entirely
Reduced 10%	Rate	–£343m	splits (2nd notch)
Reduced 15%	Rate	–£171m	splits (2nd notch)



Behavioural layer (conditional)

Layering an assumed response on the static costs.

Conditional on an **assumed** turnover elasticity e (*not* identified on synthetic data); sweep $e \in \{0.05, 0.17, 0.32\}$, midpoint 0.17. At $e = 0.17$: raise-to-£100k £508m→**£292m**; 10% band £343m→**£273m**; 15% band £171m→**£135m**, as firms broaden the base.



Conclusions

- Prices level + shape + rate on one basis; reproduces HMRC's anchor costing.
- On the dominated-region metric, **only the taper removes the distortion** — the level move is the dearest and least efficient.
- Synthetic bunching is **mechanical** (a transparency lesson) — no elasticity is read off it.
- Behavioural offsets are conditional on an assumed e ; the static cost and dominated region are the **elasticity-free anchors**.

Future work & contact

Identify e from administrative micro-data; welfare / MVPF accounting; sectoral incidence.

Vahid Ahmadi — PolicyEngine
github.com/PolicyEngine/
firm-microsim-paper

